Xin Liu

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Postdoctoral Researcher in Computational Optics

Department of Electrical and Computer Engineering
The University of Hong Kong
Hong Kong SAR, China

Research Profile Computational optics researcher developing physics-based models and algorithms to simulate and shape light for imaging and sensing.

Academic Appointments

04/2024 – Present **Postdoctoral Researcher** Hong Kong SAR, China
Department of Electrical and Computer Engineering, The University of Hong Kong (HKU)

01/2024 – 03/2024 **Research Assistant** Hangzhou, China
State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University (ZJU)

Education

09/2018 – 12/2023 **Ph.D. in Optical Engineering** Hangzhou, China
State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University (ZJU)
Dissertation: *Research on the principles and key issues of point spread function engineering*

09/2014 – 06/2018 **B.E. in Optoelectronic Information Science and Engineering** Guangzhou, China
School of Physics and Optoelectronics, South China University of Technology (SCUT)

Selected Publications

*equal contribution; ✉corresponding author.

[11] **Retained accuracy with reduced precision in wave propagation modeling.**
Xin Liu and Yifan Peng.
Photonics Research, 14(5), 1762–1774 (2026).

[10] **Learned off-aperture encoding for wide field-of-view RGBD imaging.**
Haoyu Wei, Xin Liu, Yuhui Liu, Qiang Fu, Wolfgang Heidrich, Edmund Y. Lam, and Yifan Peng.
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 1–13 (2025).

[9] **In situ fully vectorial tomography and pupil function retrieval of tightly focused fields.**
Xin Liu, Shijie Tu, Yiwen Hu, Yifan Peng, Yubing Han, Cuifang Kuang, Xu Liu, and Xiang Hao.
Nature Communications, 16(1), 3478 (2025).

[8] **Diffraction modeling between arbitrary planes using angular spectrum rearrangement.**
Yiwen Hu*, Xin Liu*✉, Xu Liu, and Xiang Hao✉.
Optica, 12(1), 39–45 (2025). **[Monthly Top Downloads in January]**

[7] **Modeling off-axis diffraction with the least-sampling angular spectrum method.**
Haoyu Wei*, Xin Liu*, Xiang Hao, Edmund Y. Lam, and Yifan Peng.
Optica, 10(7), 959–962 (2023). **[Monthly Top Downloads in July & August]**

[6] **Fast generation of arbitrary optical focus array.**
Xin Liu, Yiwen Hu, Shijie Tu, Cuifang Kuang, Xu Liu, and Xiang Hao.
Optics and Lasers in Engineering, 162, 107405 (2023).

[5] **Investigating deep optics model representation in affecting resolved all-in-focus image quality and depth estimation fidelity.**
Xin Liu*, Linpei Li*, Xu Liu, Xiang Hao, and Yifan Peng.
Optics Express, 30(20), 36973–36984 (2022).

[4] **Calibration of phase-only liquid-crystal spatial light modulators by diffractogram analysis.**
Xin Liu, Shijie Tu, Cuifang Kuang, Xu Liu, and Xiang Hao.
Optics and Lasers in Engineering, 156, 107056 (2022).

[3] **Accurate background reduction in adaptive optical three-dimensional stimulated emission depletion nanoscopy by dynamic phase switching.**
Shijie Tu, Xin Liu, Difu Yuan, Wenli Tao, Yubing Han, Yan Shi, Yanghui Li, Cuifang Kuang, Xu Liu, Yufeng Yao, Yesheng Xu, and Xiang Hao.
ACS Photonics, 9(12), 3863–3868 (2022). **[On the Cover]**

[2] **Generation of arbitrary longitudinal polarization vortices by pupil function manipulation.**
Xin Liu, Yifan Peng, Shijie Tu, Jun Guan, Cuifang Kuang, Xu Liu, and Xiang Hao.
Advanced Photonics Research, 2(1), 2000087 (2021). **[On the Cover]**

[1] **Aberrations in structured illumination microscopy: A theoretical analysis.**
Xin Liu, Shijie Tu, Yan Xu, Hongya Song, Wenjie Liu, Qiulan Liu, Cuifang Kuang, Xu Liu, and Xiang Hao.
Frontiers in Physics, 7, 254 (2020).

Awards and Honors

2025 **Outstanding Young Rising Star Award (Ranked 1st of 20)**
The 5th International Conference on Computational Imaging (CITA 2025)
Chinese Society for Optical Engineering

2024 **Honorable Mention**
The 4th International Conference on Computational Imaging (CITA 2024)
Chinese Society for Optical Engineering

2022 **Scholarship for Innovation and Entrepreneurship**
Zhejiang University

2021, 2022 **Merit Graduate Student**
Zhejiang University

2020 – 2022 **Junshi Chen Scholarship**
Zhejiang University
Awarded in 2020, 2021, and 2022.

2020, 2021 **Outstanding Graduate Student Cadre**
Zhejiang University

2019 – 2022 **Outstanding Graduate Student**
Zhejiang University
Awarded in 2019, 2021, and 2022.

Invited Talks

09/2025 **Exploring advanced imaging with computational optics** — Suzhou, China
The 5th International Conference on Computational Imaging (CITA 2025)

09/2024 **Free-space wave propagation modeling with the least-sampling angular spectrum method** — Xiamen, China
The 4th International Conference on Computational Imaging (CITA 2024)

05/2024 **Exploring wave propagation modeling in free space** — Thuwal, Saudi Arabia
Guest lecture, King Abdullah University of Science and Technology

Research Mentoring

2022 – 2025 **Learned wide-angle imaging with optical modeling and optimization** — Ph.D. thesis
Haoyu Wei
The University of Hong Kong

2021 – 2025 **Characterization of optical fields with computational methods** Ph.D. thesis
Yiwen Hu
Zhejiang University

2023 **Deep stereo optics imaging for high-fidelity depth estimation** Master's thesis
Yuhui Liu
The University of Hong Kong

2022 **Full vectorial manipulation of tightly focused optical fields** Bachelor's thesis
Haiwei Wang
Zhejiang University

2022 **Numerical simulation of vectorial fields based on vectorial diffraction theory** Bachelor's thesis
Yunpeng Wang
Changchun University of Science and Technology

Academic Service

Journal Reviewer Optica; Advanced Functional Materials; Laser & Photonics Reviews; Advanced Science; Optics Letters; Optics Express; JOSA A; JOSA B; Applied Optics; Journal of Microscopy; Optics Communications

Conference Reviewer SIGGRAPH; SIGGRAPH Asia; International Conference on Computational Photography (ICCP)

Guest Editor Computational Methods for Advanced Optical Imaging, Photonics, MDPI, 2026

Program Chair HKU-KAUST Joint Postgraduate Workshop on Computational Imaging, 2025

Program Committee International Conference on Computational Photography (ICCP), 2026
The 5th International Conference on Computational Imaging (CITA), 2025

Patents

2022 **A method and device for calibration of phase-only spatial light modulators**
Xiang Hao, Xin Liu, Yubing Han, Cuifang Kuang, and Xu Liu
Patent No. CN112697401A

2020 **A method for point spread function reconstruction**
Xiang Hao, Xin Liu, Cuifang Kuang, and Xu Liu
Patent No. CN110133849B

2020 **A two-dimensional rapid beam scanning module for confocal fluorescence microscopy**
Xiang Hao, Xin Liu, Shijie Tu, and Xu Liu
Patent No. CN109491065A

2020 **A device for common-path beam modulation in imaging and lithography systems**
Xu Liu, Xin Liu, Xiang Hao, Cuifang Kuang, and Haifeng Li
Patent No. CN110568650B

2020 **A device for generating super-resolution optical focus arrays**
Xiang Hao, Xin Liu, Cuifang Kuang, and Xu Liu
Patent No. CN110568731B

2020 **A multicolor super-resolution microscope with automatic alignment capability**
Xiang Hao, Shijie Tu, Xin Liu, and Xu Liu
Patent No. CN109358030B